

Reviewer Pack — Minimal Rank of p-Adic Valuations over Frege Proofs

Entry e2141cf11087 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of p-Adic Valuations over Frege Proofs

Entry ID: e2141cf11087 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-27 10:35:52 UTC

1. Conjecture

Field A (mathematical branch): p-adic Analysis **Field B** (complexity object): Complexity Theory: Frege Proof Complexity

Statement:

{‘quantitative_relation’: ‘The minimal rank of the p-adic valuation vectors of clauses in a Frege proof is polynomially related to the depth of the proof.’, ‘counterexample’: ‘For any fixed prime p , there exists a Frege proof with an arbitrarily large depth for which the minimal rank of the p-adic valuation vectors is constant.’}

Rationale (proposer’s reasoning):

{‘explanation’: ‘p-adic analysis provides a framework to study the properties of numbers in a way that can be applied to logical structures. The conjecture suggests that this approach could uncover new insights into the complexity of Frege proofs.’, ‘novelty’: ‘This linkage between p-adic valuations and proof complexity has not been extensively explored before, making it an underexplored area with potential for novel results.’}

Taxonomy category: p_adic_analysis_frege_proofs (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): fe6f4ab35f5821a1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of p-adic valuation vectors of clauses in a Frege proof is polynomially related to its depth if, for all generated CNFs with varying depths and clause counts, the ratio between the minimal rank and the depth is less than or equal to 2.0.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 4 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - p-adic analysis AND Frege proof complexity - minimal rank p-adic valuation AND Frege proof depth - polynomial relation p-adic valuation Frege proof

Top relevant hits considered: - [<http://arxiv.org/abs/math-ph/0512018v2>] On Phase Transitions for P -Adic Potts Model with Competing Interactions on a Cayley Tree - [<http://arxiv.org/abs/2408.00810v3>] p-adic Equiangular Lines and p-adic van Lint-Seidel Relative Bound - [<http://arxiv.org/abs/2201.08874v2>] Note on p -adic Local Functional Equation - [<http://arxiv.org/abs/hep-th/9410058v3>] p-Adic description of Higgs mechanism I: p-Adic square root and p-adic light cone

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    # Generate a random CNF with depth d and clause count c
    def generate_cnf(depth, clauses):
        variables = set()
        cnf = []
        for _ in range(clauses):
            clause = []
            for _ in range(random.randint(1, 3)):
                var = f'x{random.randint(1, variables)}'
                if random.choice([True, False]):
                    var = f'~{var}'
                clause.append(var)
                variables.add(abs(int(var[1:])))
            cnf.append(clause)
        return cnf

    # Construct a Frege proof for the given CNF
    def construct_frege_proof(cnf):
        proof = []
        for clause in cnf:
            if len(clause) == 1:
                proof.append((clause, 'axiom'))
            else:
                subproofs = [construct_frege_proof([subclause]) for subclause in clause]
```

```

        proof.append((clause, 'resolution', *subproofs))
    return proof

# Compute the p-adic valuation vector of a clause
def p_adic_valuation_vector(clause):
    p = 2 # Using base 2 for simplicity
    vector = [0] * (len(variables) + 1)
    for var in clause:
        if var.startswith('-'):
            var = var[1:]
            sign = -1
        else:
            sign = 1
        index = int(var[1:])
        vector[index] += sign
    return vector

# Compute the minimal rank of a list of vectors
def min_rank(vectors):
    if not vectors:
        return 0
    n, m = len(vectors), len(vectors[0])
    rank = 0
    for i in range(m):
        pivot_found = False
        for j in range(rank, n):
            if vectors[j][i] != 0:
                vectors[j], vectors[rank] = vectors[rank], vectors[j]
                pivot_found = True
                break
        if not pivot_found:
            continue
        rank += 1
        for j in range(n):
            if j != rank - 1 and vectors[j][i] != 0:
                factor = vectors[j][i] / vectors[rank - 1][i]
                for k in range(m):
                    vectors[j][k] -= factor * vectors[rank - 1][k]
    return rank

# Generate a random CNF with depth d and clause count c
depth = random.randint(5, 40)
clauses = random.randint(20, 80)
cnf = generate_cnf(depth, clauses)

# Construct a Frege proof for the given CNF
proof = construct_frege_proof(cnf)

# Compute the p-adic valuation vectors of all clauses in the proof
valuation_vectors = []
for clause, _, *subproofs in proof:
    valuation_vector = p_adic_valuation_vector(clause)
    valuation_vectors.extend(valuation_vector for _ in subproofs)

# Compute the minimal rank of the p-adic valuation vectors
min_rank_value = min_rank(valuation_vectors)

```

```

# Measure the depth of the Frege proof
depth_value = len(proof)

# Check if the conjecture holds
conjecture_holds = min_rank_value <= 2.0 * depth_value

return {
    "metric_name": "min_rank_to_depth_ratio",
    "metric_value": min_rank_value / depth_value,
    "instances_tested": len(valuation_vectors),
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else f"Depth={depth}, Min Rank={min_rank_value}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, **{result}}}")
        results.append(result)

    mean_d = sum(r["metric_value"] for r in results) / len(results)
    std_dev = math.sqrt(sum((r["metric_value"] - mean_d) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_d} std={std_dev} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_d} std={std_dev} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        counterexample = next((r["counterexample"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5d62f30a.py", line 121, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5d62f30a.py", line 87, in run_trial
    cnf = generate_cnf(depth, clauses)
          ^^^^^^^^^^^^^^^^^^^^^^^^^

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_5d62f30a.py", line 28, in generate_cnf
    var = f'x{random.randint(1, variables)}'
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

```

```
File "/usr/lib/python3.12/random.py", line 336, in randint
    return self.randrange(a, b+1)
           ~^~
```

TypeError: unsupported operand type(s) for +: 'set' and 'int'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying the conjecture's conditions. | next: Investigate and fix the error in the test code to allow for proper verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	10764
2	preregistration	ollama_remote	glm4:latest	0	0	5648
3	novelty	ollama_remote	glm4:latest	0	0	4505
4	novelty	ollama_remote	glm4:latest	0	0	6217
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12509
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7646
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7862
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13757
9	judge	ollama_remote	glm4:latest	0	0	8236

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 77143 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/e2141cf11087.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/e2141cf11087.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/e2141cf11087.tar.gz` (if generated)